📊 Mr_Ed' Sampling Suite

A Premium, Offline-First Scientific Survey Sampling, Calibration, Weighting, and Variance Estimation Suite for Official Statistics.

React 19.0
TypeScript 5.x
Vite 8.x
TailwindCSS 3.x
Security GDPR / HIPAA Compliant

🌟 Overview

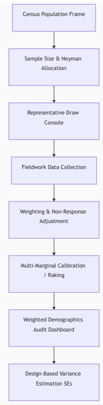
Mr_Ed' Sampling Suite is an advanced, high-performance web and desktop-wrapped dashboard designed for national statistics offices, survey researchers, and official statisticians. It offers professional, mathematically rigorous, and end-to-end control of the survey lifecycle—including **sample size planning**, **random sample drawing**, **non-response adjustment**, **multi-marginal demographic calibration (raking)**, and **stratified design-based variance estimation**—all executing completely locally on your computer with zero external database dependencies.

🔒 GDPR Secure & Offline-First

Unlike online AI tools or cloud-based data portals, **Mr_Ed' Sampling Suite runs 100% client-side inside local browser memory**.

- **Zero Uploads:** Your census directories, population frames, and sensitive survey respondent records are never uploaded to any remote server or third-party cloud.
- **National Security Compliant:** Operates seamlessly on air-gapped computers, behind secure government firewalls, or inside national statistics security networks.
- **GDPR & HIPAA Compliant:** Safeguards personally identifiable information (PII) by design.

🚀 Key Modules & Capabilities



1. Executive Dashboard (System Control Center)

- **High-Performance Upload:** Instantly parse million-row census population files using an optimized client-side engine.
- **Global Stats Ribbon:** Tracks population frame parameters, active sample quotas, and calibrated response cases.
- **Dynamic Sidebar Navigation:** Seamlessly jump across the 6 major workflow modules using a premium glassmorphic interface.

2. Sample Size & Stratum Allocation Suite

- **Interactive Math Sandbox:** Calculate target sizes in real-time by dragging sliders for confidence levels ($1 - \alpha$), margins of error (e), expected proportions (p), or continuous standard deviations (S).
- **Flexible Estimators:** Toggle between **Cochran's Proportions Formula**, **Cochran's Continuous Mean**, and **Slovin's (Yamane)** equations.
- **Finite Population Correction (FPC):** Automatically detects if the sampling fraction exceeds 5% and compresses the variance dynamically.
- **Quota Allocation:** Distribute baseline sample sizes across geographic or demographic strata using **Proportional** or **Neyman Optimal Allocation** (which minimizes variance by targeting high-dispersion subgroups).

3. Interactive Methodology & Knowledge Hub

- **Methodology Timeline:** A visual, vertical lifecycle detailing the mathematical rationale, caveats, and R/SAS equivalence of every survey step.
- **Timed Grid Visualizer:** A 10×10 CSS-animated grid displaying unweighted vs. weighted unit drawing patterns for **SRS**, **Systematic**, **Stratified**, and **Cluster** designs.

- **Kish's Design Effect (Deff) Calculator:** Real-time dials simulating variance inflation factors due to clustering ($\mathrm{Deff} = 1 + (m-1)\rho$) or unequal weighting dispersion ($\mathrm{Deff} = 1 + \mathrm{CV}(w)^2$).
- **Step-by-Step Raking IPF Solver:** A 2×2 dynamic table demonstrating how Iterative Proportional Fitting converges weights to demographics.

4. Advanced Weighting & Calibration Suite

- **Flexible Data Channels:** Calibrate weights using either the actively drawn sample or by uploading a external fieldwork survey dataset.
- **Non-Response Modeling:** Correct for demographic attrition using **Weighting Class Adjustments** or **Response Propensity Logistic Regressions**.
- **Three Calibration Engines:**
  1. **Multiplicative Raking (Ratio/IPF):** Fits multi-marginal demographics iteratively while keeping weight multipliers positive.
  2. **Linear Calibration (GREG Solver):** Uses single-step Lagrange equations equivalent to Generalized Regression estimators.
  3. **Logit Calibration:** Enforces strict boundary thresholds ($L \leq w_i/d_i \leq U$) to prevent excessive weight inflation.

5. Weighted Demographics & Representativeness Dashboard

- **Dissimilarity Index (D):** Compares unweighted vs. weighted distributions against target census margins.
- **Multi-Variable Chart Grid:** Beautiful comparative charts tracking category-level representativeness before and after weights calibration.
- **Audit Exports:** Export structured `.xlsx` demographic reports detailing counts, weighted percentages, target deviations, and average multiplier scales.

6. Variance & Analytics Engine

- **Stratified Taylor Series Linearization:** Computes design-based standard errors matching standard packages like R's `survey` or SAS's `PROC SURVEYMEANS`.
- **Bootstrap Replicas Resampling:** McCarthy-Snowden or Rao-Wu cluster bootstrap weight generation (e.g., 100 replicates).
- **Calibration Re-Estimation:** Re-calibrates every single replicate column during bootstrap iterations to capture calibration variance reductions perfectly.

🛠️ Technology Stack

- **Core Framework:** React 19.0, Vite 8.0, TypeScript 5.x
- **Styling & UI:** TailwindCSS 3.x, Lucide Icons, Glassmorphic space-mesh aesthetic
- **Excel Engine:** SheetJS (`xlsx`) for local client-side spreadsheet parsing and generation
- **Data Visualizations:** Chart.js + React-Chartjs-2 for high-resolution graphics
- **Math Notation:** LaTeX parsing for mathematical formula sandboxes

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📦 Installation & Local Setup

To run the application locally on your computer:

Prerequisites

Make sure you have Node.js (v18 or higher) installed.

Step 1: Clone the Codebase

```
git clone https://github.com/YOUR_USERNAME/mr-eds-sampling-suite.git
cd "mr-eds-sampling-suite"
```

Step 2: Install Dependencies

```
npm install
```

Step 3: Run the Development Server

```
npm run dev
```

Open http://localhost:5173 in your browser to view the application running locally!

Step 4: Build for Production

To compile and package the app for web deployment:

```
npm run build
```

This generates a highly optimized, minified static site bundle inside the `/dist` directory.

🚀 Instant Web Deployment

To host this application online completely for free:

Deploy with Vercel CLI (Takes 2 minutes)

1. Install the Vercel command line interface globally:

   ```
   npm install -g vercel
   ```

2. Run the deployment command in the root folder:

   ```
   vercel
   ```

3. Follow the interactive CLI prompts. Your application will be live at `https://mr-eds-sampling-suite.vercel.app`!

📄 License

This project is licensed under a custom Enterprise Non-Distribution License. All core computation libraries, raking algorithms, and mathematical models operate locally and are fully owned by the workspace administrator.